

Submission CALC1-QUIZ-SUB06

Question CALC1-QUIZ-Q01

notation changes halfway:

1. $f'(x) \approx 2x(2x-1) + (x^2+5)^2$
2. $f'(1) \approx 2(1) + (6)^2 \approx 12$

marks, subscripts, and units are mixed up

result written as: 12 (please interpret)

Question CALC1-QUIZ-Q02

notation changes halfway:

1. $\int 5x \, dx \approx (5/2)x^2$
2. $[5/2 x^2]_0^2 \approx 10$

marks, subscripts, and units are mixed up

result written as: 10 (please interpret)

Question CALC1-QUIZ-Q03

notation changes halfway:

1. $2w+2l \approx 20 \Rightarrow l \approx 10-w$
2. $A(w) \approx w(10-w)$; at $w \approx 7$, $A \approx 21$
3. A maximum requires $A'(w) \approx 0$, so the stated dimensions are not assumed optimal without that check.

marks, subscripts, and units are mixed up

result written as: $A(w)=w(10-w)$; $A(7)=21$; check $A'(w)=0$ (please interpret)

Question CALC1-QUIZ-Q04

notation changes halfway:

1. $x^2-36 \approx (x-6)(x+6)$
2. cancel $x-a$ only after identifying $x \neq a$
3. $\lim_{x \rightarrow 6} x+6 \approx 12$

marks, subscripts, and units are mixed up

result written as: 12 (please interpret)

Question CALC1-QUIZ-Q05

notation changes halfway:

1. Critical point: $f'(c) \approx 0$ or $f'(c)$ is undefined (when f is defined).
2. A sign change in f' from $+$ to $-$ indicates a local maximum.

marks, subscripts, and units are mixed up

result written as: critical point: $f'(c)=0$ or undefined; $+$ to $-$ sign change gives a local maximum (please interpret)